

TrES-5b

R-0991

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_200925 · created 2026-07-29T23:24:14+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459117.7375 +/- 0.0032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.183 +/- 0.052
Transit depth (Rp/Rs)^2	3.35 +/- 1.91 [%]
Orbital Inclination (inc)	85.11 +/- 2.55
Airmass coefficient 1 (a1)	1.307 +/- 0.042
Airmass coefficient 2 (a2)	-0.192 +/- 0.076
Scatter in the residuals of the lightcurve fit is	3.19 %
Best Comparison Star	None
Optimal Aperture	2.62
Optimal Annulus	5.01
Transit Duration (day)	0.0586 +/- 0.0091

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.183 $\pm$ 0.052 vs 0.142 (deviation 0.56 σ)
Duration fit vs published	0.0586 d vs 0.0758 d (deviation 1.34 σ)
Mid-time O-C	-7.3 min
Depth SNR	2.5
Residual scatter	3.19 %

Light curve

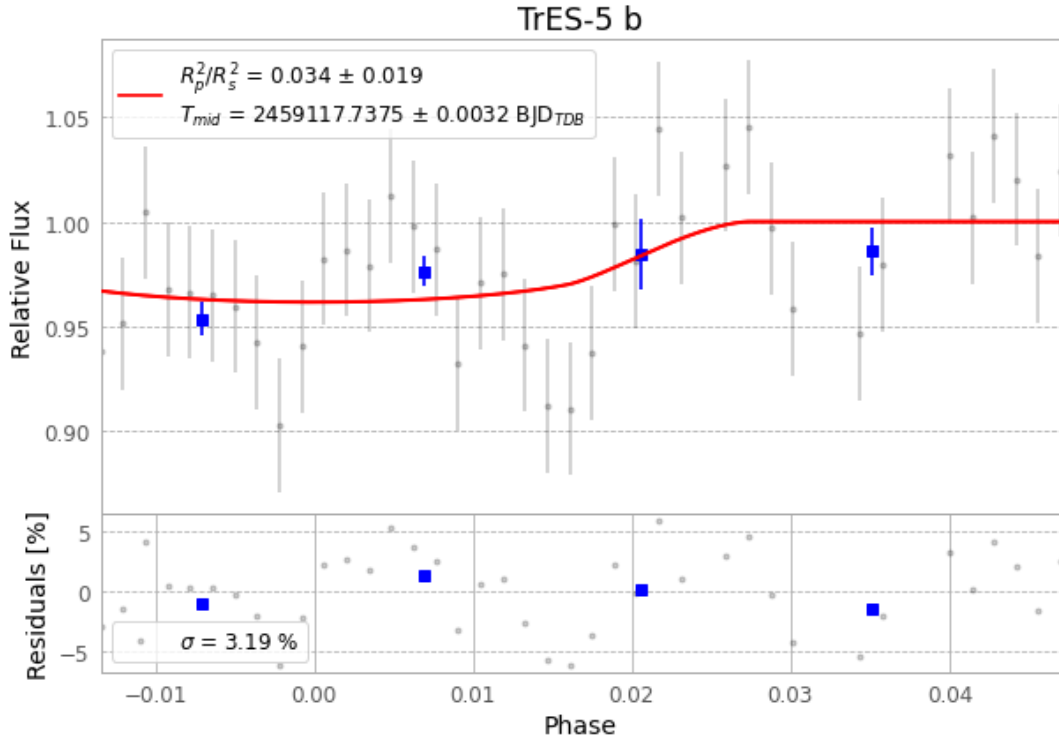


Figure 1 — FinalLightCurve_TrES-5 b_2020-09-24.png (SHA-256 f9cc2b9e6ab00c92...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [413, 353], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 bf8a1c26a37f81b2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

40 FITS frames (26 MB), TRES-5200925050912.FITS → TRES-5200925071812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-200925.log (2b16d8698488...), FinalLightCurve_TrES-5 b_2020-09-24.png (f9cc2b9e6ab0...), FinalLightCurve_TrES-5 b_2020-09-24.csv (a77a92cc807c...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 23:24Z.